

Quick Sort

Definition

Quick Sort is a divide-and-conquer sorting algorithm. It selects an element called a **pivot**, partitions the array around the pivot, and then recursively sorts the smaller sub-arrays.

Algorithm

1. Choose an element as the **pivot**.
2. Partition the array so that:
 - a. Elements smaller than the pivot are on the left.
 - b. Elements greater than the pivot are on the right.
3. Place the pivot in its correct position.
4. Apply Quick Sort recursively to the left sub-array.
5. Apply Quick Sort recursively to the right sub-array.
6. Continue until the sub-arrays contain one or zero elements.

Output:

```
Enter number of elements: 5
Enter element: 4
Enter element: 9
Enter element: 6
Enter element: 3
Enter element: 2
Original array: [4, 9, 6, 3, 2]
Sorted array: [2, 3, 4, 6, 9]
Time Complexity:
Best Case:  $O(n \log n)$ 
Average Case:  $O(n \log n)$ 
Worst Case:  $O(n^2)$ 
Space Complexity:  $O(n \log n)$ 

...Program finished with exit code 0
Press ENTER to exit console.
```

Space Complexity

- **Average:** $O(\log n)$ — recursion stack
- **Worst:** $O(n)$ — highly unbalanced recursion

Final Answer

Best Time = $O(n \log n)$

Average Time = $O(n \log n)$

Worst Time = $O(n^2)$

Space = $O(\log n)$ average, $O(n)$ worst case